Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/28/2005
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Project Title:	Test of the Equivalence Principle in an Einstein Elevator		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Glashow, Sheldon (Boston University Boston MA) Iafolla, Valerio (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Lorenzini, Enrico (Smithsonian Astrophysical Observatory Cambridge MA) Cheimets, Peter (Smithsonian Astrophysical Observatory Cambridge MA) Cosmo, Mario (Smithsonian Astrophysical Observatory Cambridge MA) Finkelstein, Nancy (EP/O) (Smithsonian Astrophysical Observatory Cambridge MA) Schneps, Matthew (EP/O) (Smithsonian Astrophysical Observatory Cambridge MA) Nozzoli, Sergio (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Santoli, Francesco (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Lucchesi, Davide (Institute of Interplanetary Space Physics (IFSI) Rome Italy)		
Grant/Contract No.:	NAG3-2881		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics anbd unlock the mysteries of the universe obscured by gravity		

Task Description:	The goal of this grant is to carry out the experiment definition of a system for testing repeatedly the Equivalence Principle with an accuracy significantly improved with respect to the state of the art. The experimental technique utilizes a differential accelerometer (detector) in free fall inside an evacuated capsule released from a stratospheric balloon. The detector free falls for as long as 30 seconds, while rotating slowly, inside the co-moving shielding capsule which drops up to 4.4 km while providing a very high level of isolation from external noise sources. The detector measures very accurately the separation (and hence the differential rate of fall) of a pair of test masses of different composition. The experiment can be repeated periodically and, consequently, allow for fine tuning of the experiment, improvements of the detector, and testing of different materials. The ultimate objective of the experiment is to test the principle that lies at the foundation of the currently-accepted gravitational theory (general relativity), that is, the equivalence between the mass seen as a response to gravity and the mass seen as a repository of inertia. The goal of the flight experiments is to push the accuracy in testing this equivalence up to a few parts in 10^15 or, similarly, in measuring the relative acceleration in free fall of a pair of test masses of different composition (the sensing elements of the differential accelerometer) up to that accuracy. The target accuracy is about two orders of magnitude better than the present state of the art. The experiment definition activities covered by this grant will be as follows: (a) assessment of science requirements and definition of engineering requirements for the experiment; (b) analysis of noise sources and their mitigation; (c) preliminary design of the free-fall system and its acceleration detector for the balloon-flight experiments; (d) construction and laboratory testing of a prototype release mechanism; (e) analysis of the theoretical implications of testing the Equivalence Principle with an improved accuracy; (f) development of techniques for extracting a small signal from data strings with dominant noise; and (g) an E/PO activity with the goal of conveying basic ideas of gravitational physics to students and educators. In parallel to the tasks above, the group at the Institute of Space Physics (IFSI) in Rome, funded by the Italian Space Agency, will develop prototype differential accelerometers and conduct key tests on the prototype accelerometers in the laboratory. The Equivalence Principle is at the foundation of gravitational theories and thus affects our understanding of the fundamental forces in nature. A negative result of the test (non violation) will impose very stringent limits on gravitational theories while a positive result (violation) will likely imply the need for a fundamental revision in the theory of general relativity and, indeed, in other basic laws developed to describe nature. The results of the tasks to be carried out under this grant will define the scientific and technical requirements of the experiment, assess its scientific/technical soundness, and lay the groundwork for the balloon experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The research activity during this grant year has focused on the following tasks: (a) analysis and conceptual design of a differential acceleration detector that satisfy the requirements of the balloon-flight experiments; (b) development of techniques for extracting a mono-chromatic, small signal from data strings affected by dominant noise; (c) design of the mechanism that spins and releases the instrument package (housing the detector) inside the falling capsule/cryostat; and (d) construction of a new differential accelerometer prototype by our non-US partners and its preliminary testing in the laboratory. This grant-year accomplishments are briefly summarized in the following. The analysis of various configurations for the balloon-flight detector and the simulation of their dynamic response has yielded a conceptual design of a detector that provides: (1) an excellent frequency separation between the possible Equivalnce Principle violation signal and the dynamics-related noise (inclusive of gravity gradients); and (2) a violation signal that is detectable at the threshold sensitivity level of the detector. In other words, with this new design the ability to detect a small signal is not limited by the dynamics-related noise and gravity gradients but rather by the intrinsic noise of the detector and its electronics (e.g., Brownian and preamplifier noise). Moreover, we have developed techniques for extracting a signal at the threshold level from data strings affected by the much stronger, colored, dynamics-related noise and the intrinsic white noise of the detector. We have tested those techniques on the simulated output data of various detector's configurations. We have computed the probability of detecting a small signal (out of white noise) and the likely uncertainty in estimating its amplitude as a function of the signal to noise ratio. Our engineering department has designed a spin and release mechanism, consisting of two separation stages, and identified the key mechanical and electrical components. Our non-US partners have built a second differential accelerometer prototype that has several features in common with the balloon-experiment detector. Our non-US partners also carried out preliminary tests in the laboratory of the new differential accelerometer prototype.
Bibliography Type:	Description: (Last Updated: 07/09/2007)
Articles in Peer-reviewed Journals	Iafolla, V.; D.M. Lucchesi, S. Nozzoli; M. Ravenna; F. Santoli; I.I. Shapiro; E.C. Lorenzini; M.L. Cosmo; C. Bombardelli; J. Ashenberg; P.N. Cheimets; S. Glashow. "General Relativity Accuracy Test (GReAT): new configuration for the differential accelerometer." Submitted to Advances in Space Research. , Oct-2004
Papers from Meeting Proceedings	Lorenzini, E.C.; I.I. Shapiro; M.L. Cosmo; P.N. Cheimets; C. Bombardelli; V. Iafolla; D.M. Lucchesi; S. Nozzoli; M. Ravenna; F. Santoli; S. Glashow. "Development of the TEPEE/GReAT experiment to test the Equivalence Principle in Free Fall from a balloon." Conference on Testing the Equivalence Principle in Space and on Ground, Pescara, Italy, 20-23 September 2004. Sep-2004
Papers from Meeting Proceedings	Iafolla, V.; D.M. Lucchesi; S. Nozzoli; M. Ravenna; F. Santoli; I.I. Shapiro; E.C. Lorenzini; M.L. Cosmo; J. Ashenberg; P.N. Cheimets; S. Glashow. "The Weak Equivalence Principle (WEP) and the General Relativity Accuracy Test (GReAT) with an Einstein Elevator." 16th SIGRAV Conference on General Relativity and Gravitational Physics, Vietri, Italy, 13-16 September 2004. Sep-2004
Presentation	Iafolla, V.; D.M. Lucchesi; S. Nozzoli; M. Ravenna; F. Santoli, I.I. Shapiro; E.C. Lorenzini; M.L. Cosmo; C. Bombardelli; J. Ashenberg; P.N. Cheimets; S. Glashow. "General Relativity Accuracy Test (GReAT): new configuration for the differential accelerometer." Presented at the COSPAR-2004 Congress, Paris, France, July 2004. Jul-2004